Answer Key & Teacher Notes

Session 1 — Mean or Median? · Session 2 — Matching Center and Spread

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Mean or Median?

TALK PROMPTS

- How can one unusual value change the mean without changing the median?
- For fares $12, $14, $16, $18, $95, which of your two numbers would a rider recognize as a usual fare, and why?

PARTNER TASK

Movie Runtimes. One partner finds the median of 95, 98, 101, 103, 143; the other finds the mean. Together decide which one the long film pulls and which better describes a typical movie.

WATCH FOR

- Choosing the mean for data with an outlier because it "uses every value."
- Finding the median without first putting the values in order.
- With an even count, picking one middle value instead of averaging the two middle values.

SESSION 2 · TEACHER NOTES

Matching Center and Spread

TALK PROMPTS

- Why do the mean and the MAD belong together, and the median and the IQR belong together?
- In Problem 7, what single feature of the new value, 500, changed your choice of pair?

PARTNER TASK

Two Sets in Problem 1. One partner names the shape and pair for 1a (31, 33, 35, 37, 39); the other for 1b (31, 33, 35, 37, 290). Then each justifies why 1b needs a different pair.

WATCH FOR

- Mixing the pairs (median with MAD, or mean with IQR).
- Missing a low outlier (skewed left) because no value is far above the rest.
- Reading a long right tail as skewed left.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.5d	1	2	2	1	2	2	2	2	2	1	1
S1 · Form B	6.SP.B.5d	1	2	2	1	2	2	2	2	2	1	1
S2 · Form A	6.SP.B.5d	1	2	2	2	2	2	2	2	2	1	1
S2 · Form B	6.SP.B.5d	1	2	2	2	2	2	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

outlier: A value **far** away from the others.

skewed: Most values sit on one side, with a long **tail**.

symmetric: Both sides of the data look about the **same**.

measure of center: One number that shows a **typical** value.

- An outlier pulls the **mean** toward it, but barely moves the median.
- Skewed data or an outlier: use the **median**. Symmetric data: use the mean.

Example: The outlier is **50**. Median: the middle of 6, 7, 8, 9, 50 is **8**. Mean: 80 ÷ 5 = **16**, more than four of the five savings. Answer: Use the **median**: about $8.

YOU TRY

1. a. mean (symmetric) b. median (65 is an outlier)

2. mean 75 ÷ 5 = $15; median $6; the median ($53 is an outlier)

EXIT TICKET

1. a. mean (no outlier) b. median (65 is an outlier)

2. (20 + 22) ÷ 2 = 21 min (the mean, 50 min, is pulled up by 200)

3. **B** (mean)

PRACTICE FORM A

1. a. mean (symmetric) b. median (500 is an outlier) c. mean (symmetric) d. median (900 is an outlier)

2. a. 88 b. 23 years c. 175 ÷ 5 = 35 years d. the median; 35 is greater than four of the five ages

3. a. 101 min b. 540 ÷ 5 = 108 min c. the mean (pulled up by 143)

4. a. $16 b. 155 ÷ 5 = $31

5. No. Mean = 420 ÷ 5 = 84, more than four of the five values. The median, 57, is typical.

6. a. 420 ÷ 6 = 70 b. (42 + 44) ÷ 2 = 43 c. the median; 210 is an outlier that pulls the mean above five of the six salaries

7. mean 400 ÷ 5 = 80; median 80; either works because the set is symmetric (report the mean)

8. a. 12 b. 140 ÷ 7 = 20 c. the median (72 is an outlier); pair it with the IQR

9. 148 is an outlier that pulls the mean up; only one value is near 33. He should report the median, 5.

10. **C** (median)

11. **A** (IQR)

Self-check word **OUTLIER**

PRACTICE FORM B

1. a. mean (symmetric) b. median (700 is an outlier) c. mean (symmetric) d. median (1 is an outlier)

2. a. 78 b. 19 years c. 150 ÷ 5 = 30 years d. the median; 30 is greater than four of the five ages

3. a. 99 min b. 545 ÷ 5 = 109 min c. the mean (pulled up by 154)

4. a. $15 b. 140 ÷ 5 = $28

5. No. Mean = 360 ÷ 5 = 72, more than four of the five values. The median, 47, is typical.

6. a. 408 ÷ 6 = 68 b. (40 + 42) ÷ 2 = 41 c. the median; 208 is an outlier that pulls the mean above five of the six salaries

7. mean 370 ÷ 5 = 74; median 74; either works because the set is symmetric (report the mean)

8. a. 13 b. 140 ÷ 7 = 20 c. the median (72 is an outlier); pair it with the IQR

9. 154 is an outlier that pulls the mean up; only one value is near 35. He should report the median, 6.

10. **D** (mean)

11. **C** (MAD)

Self-check word **MEDIANS**

GUIDED NOTES

symmetric: Both sides of the data look about the **same**.

skewed: Most values sit on one side, with a long **tail**.

mean and MAD: The pair for **symmetric** data with no outlier.

median and IQR: The pair for skewed data or data with an **outlier**.

- MAD is built from the **mean**; IQR is built from the median. Never mix the pairs.
- Look for an outlier or a long **tail** before you choose a pair.

Example: Set C is **symmetric**: mean 24, MAD 12 ÷ 5 = 2.4. Set D has an outlier, **150**: median 24. Answer: C: mean and MAD. D: **median** and IQR.

YOU TRY

1. a. mean and MAD b. median and IQR

2. median 28; Q1 = 24, Q3 = 32; IQR = 8

EXIT TICKET

1. a. mean and MAD b. median and IQR

2. median 9; Q1 = 6, Q3 = 12; IQR = 6

3. **C** (median with MAD)

PRACTICE FORM A

1. a. mean and MAD b. median and IQR c. mean and MAD d. median and IQR

2. a. skewed right (long tail to the right) b. median and IQR

3. a. 280 ÷ 5 = 56 b. distances 4, 2, 0, 2, 4; MAD = 12 ÷ 5 = 2.4

4. a. 17 b. Q1 = 13, Q3 = 21, IQR = 8

5. The IQR. The MAD is built from the mean, so it does not match the median.

6. symmetric, so mean and MAD; mean = 217 ÷ 7 = 31

7. a. mean and MAD b. median and IQR (500 is an outlier) c. (84 + 86) ÷ 2 = 85

8. a. skewed right (long right whisker) b. 15 min c. 30 − 10 = 20 min

9. Using every value is the problem: the outlier pulls the mean and the MAD. With an outlier he should report the median and the IQR.

10. **A** (mean and MAD)

11. **C** (mean with IQR)

Self-check word **SYMMETRY**

PRACTICE FORM B

1. a. mean and MAD b. median and IQR c. mean and MAD d. median and IQR (1 is an outlier)

2. a. skewed left (long tail to the left) b. median and IQR

3. a. 225 ÷ 5 = 45 b. distances 5, 2, 0, 2, 5; MAD = 14 ÷ 5 = 2.8

4. a. 19 b. Q1 = 16, Q3 = 24, IQR = 8

5. The MAD. The IQR is built from the median, so it does not match the mean.

6. symmetric, so mean and MAD; mean = 483 ÷ 7 = 69

7. a. mean and MAD b. median and IQR (400 is an outlier) c. (64 + 66) ÷ 2 = 65

8. a. skewed right (long right whisker) b. 35 min c. 45 − 25 = 20 min

9. Using every value is the problem: the outlier pulls the mean and the MAD. With an outlier she should report the median and the IQR.

10. **B** (skewed right)

11. **D** (median)

Self-check word **BALANCED**

SHEET 1 · MEAN OR MEDIAN?

1. a. mean (symmetric) b. median (3 is an outlier) c. median (75 is an outlier)
2. a. symmetric b. 63 ÷ 9 = 7 c. 7 d. the mean (symmetric, no outliers; mean and median are equal)
3. a. skewed right b. the mean (the few high prices in the tail pull it up) c. the median
4. a. \$78 ÷ 6 = \$13 b. (\$8 + \$8) ÷ 2 = \$8 c. the median; \$41 is an outlier, and the mean (\$13) is more than 5 of the 6 allowances
5. \$62,000 is the mean, pulled up by the one \$178,000 salary; 4 of the 5 workers earn far less. The median, \$34,000, describes typical pay better.

SHEET 2 · MATCHING CENTER AND SPREAD

1. a. mean and MAD (symmetric) b. median and IQR (95 is an outlier)
2. a. mean and MAD (symmetric) b. mean = 32 ÷ 8 = 4 books c. distances 2, 1, 1, 0, 0, 1, 1, 2; MAD = 8 ÷ 8 = 1 book
3. a. skewed right (long right whisker) b. 14 min c. 20 − 12 = 8 min
4. a. median and IQR (70 is an outlier) b. 18 min c. Q1 = 15, Q3 = 22, IQR = 7 min
5. The value 2 is a low outlier, far below the rest, so the data is skewed left. He should report the median and IQR: median 22, IQR 24 − 20 = 4.
6. **B** (median and IQR)
7. **A** (mean and MAD)

October 2026 conceptual additions — teacher key

Session 1 · Mean or Median?

Total 40, mean 8, median 4. Median 4 is a useful central-family description because the large collection raises the mean. Mean 8 gives the equal share for five boxes. The purpose matters.

Extension: The mean is correctly calculated and answers total divided by count; it may be less useful for a particular description of a typical observation.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Matching Center and Spread

A is symmetric about 8: mean 8 and MAD 2.4 are useful. B has a large high value: median 8 and IQR 16 are defensible; Q1 5, Q3 21. Do not claim the IQR is unaffected merely because it is resistant.

Extension: The upper half of B is 10,32, so Q3 depends on 32. Resistance is not immunity; compute and interpret the actual summaries.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.